Vu Xuan Hien

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SUMMARY

Machine Learning Engineer with experience building and deploying ML systems end-to-end, from large-scale data pipelines and feature engineering to model deployment, monitoring, and scaling in production. Experienced in delivering LLM-powered systems, credit-risk models, and computer vision applications across fintech, industrial, and enterprise settings. Proficient in Python, PyTorch, Spark, and AWS, with deep MLOps experience across the full model lifecycle.

WORK EXPERIENCE

Machine Learning Engineer — GE Vernova (United Kingdom) **Nov 2025 – Present**

- Built a RAG assistant over the user guides of an in-house engineering design platform, with structure-aware chunking for mixed docs (tables, code, YAML, images) and hybrid retrieval (dense + BM25) plus a reranker. Raised precision@5 from 59% to 70% and cut documentation search time by an estimated 40%
- Served it as a FastAPI service in Docker on AWS EC2 behind a load balancer with SSO, checked grounding with an LLM-as-judge faithfulness rubric. User feedback led to parsing fixes and automatic re-ingestion on doc updates
- Designed an agent on the OpenAI Agents SDK and a custom MCP server so engineers can run the platform in plain language: optimisation runs, datasets, ML workflows. A planner orchestrates each request into tool calls, and guardrails validate arguments and ask the engineer to confirm before risky actions run
- Run the team's LLM gateway (LiteLLM, Postgres, Docker) for about 15 engineers: routing across OpenAI and Gemini, API key management, per-user rate limits, cost and latency dashboards

Machine Learning Engineer — Data Nest (Fintech, Vietnam) **Apr 2024 – Sep 2024**

- Built an income-scoring model in CatBoost on telco data covering about 70% of Vietnam's population, reaching 0.78 AUC. Lenders used it to automate income checks, cutting manual review time by about 45%
- Engineered features from 10B+ daily telco records in Spark, scheduled with Airflow, lifting precision by 10%
- Caught a label leak before production: a fixed-date aggregation window captured post-default behaviour and inflated AUC, rebuilt the features with per-customer cutoff dates
- Monitored the scoring models in production with Grafana drift dashboards and caught 4 drift events, traced each to its root cause, flagged it to the business and retrained, with runs tracked in MLflow

AI Engineer — FPT Software (Vietnam) **Sep 2022 – Apr 2024**

- Developed a video understanding system for human tracking and action recognition, combining pose estimation with an LSTM on top of a pre-trained vision-language model. Reached 96% accuracy across 3 action classes on a client test set
- Designed a learning-path recommender using hybrid collaborative filtering and knowledge tracing, course completion and time on platform rose about 30% the following semester
- Trained a GAN to generate Japanese handwriting from printed text as synthetic OCR training data, raising OCR accuracy from 73% to 93%
- Integrated LangChain RAG and custom OCR models into a chatbot for Japanese document search and summarisation, deployed for a client company

EDUCATION

MSc in Artificial Intelligence and Machine Learning — University of Birmingham (With Distinction) **2024 – 2025**

Research Focus: Safety in LLM-based AI Agents

BSc in Biomedical Engineering — Vietnam National University, International University **2018 – 2022**

SKILLS

Programming: Python, C, Java, R

Machine Learning: PyTorch, TensorFlow, scikit-learn, Catboost, Transformers, feature engineering, drift monitoring

MLOps: AWS, Azure, GCP, FastAPI, Docker, Kubernetes, MLflow, DVC, Airflow, CI/CD, Git, Grafana

Data: SQL, Spark, Kafka, HDFS, PostgreSQL, MongoDB

LLM & Agents: RAG, OpenAI Agents SDK, MCP, LangChain, LangGraph, Ragas, Chroma, Pinecone

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (2024)
- Machine Learning DevOps Engineer – Udacity